

Problem Set #1: Solutions and Empirical Analysis

Macroeconomic Measurement, National Accounts Decomposition, Goods Market Equilibrium, and Multipliers

1 Real GDP and the GDP Deflator (20 Points)

(1) Nominal and Real GDP Time Series (10 points)

Annual macroeconomic data are retrieved from the Census and Statistics Department (C&SD) of the Hong Kong Special Administrative Region (Table 310-31001: *Gross Domestic Product (GDP), implicit price deflator of GDP and per capita GDP*). We extract “GDP at current market prices” (nominal GDP, $Y_t^{\$}$) and “GDP in chained (2024) dollars” (real GDP, Y_t) for the period 1961–2025 (65 annual observations). Figure 1 plots both series, constructed analogously to Figure 2-1 in Blanchard (2021).

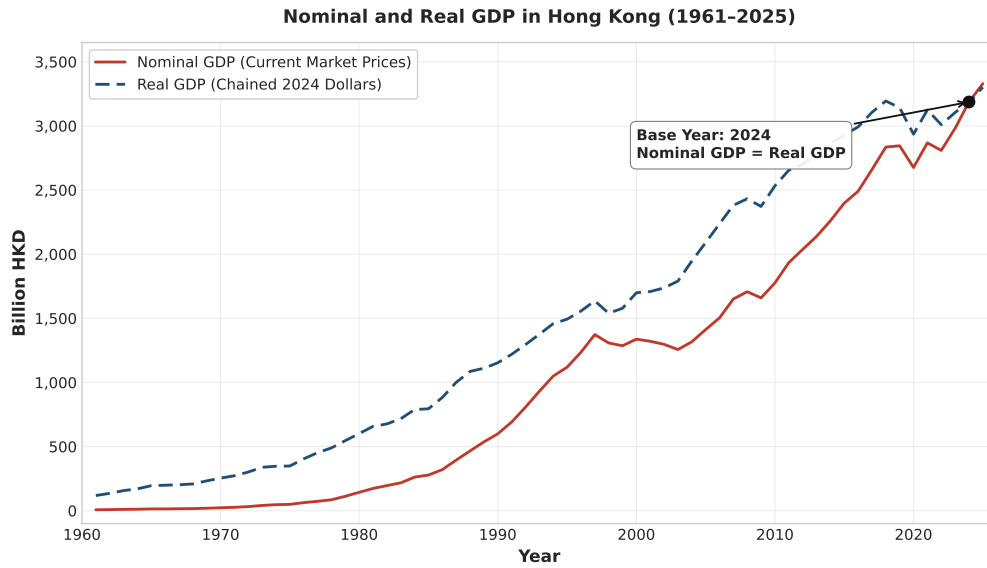


Figure 1: Nominal and Real GDP in Hong Kong (1961–2025)

Data Source: C&SD Table 310-31001. Real GDP is expressed in chained (2024) dollars with base year 2024.

Economic and Structural Interpretation:

- **Base Year Intersection:** The nominal and real GDP curves intersect precisely at the reference base year **2024**, where $\text{Nominal GDP}_{2024} = \text{Real GDP}_{2024} = \text{HK\$ } 3,186,526$ million, by definition setting the implicit price deflator $P_{2024} \equiv 100$.
- **Pre-2024 Volume Dynamics ($P_t < 100$):** Prior to 2024, real GDP lies strictly above nominal GDP. Because historical price levels were substantially lower than the 2024 base price, evaluating historical output at 2024 chained prices scales up the physical volume of production relative to its current-market-price valuation.
- **Post-2024 Trajectory ($P_t > 100$):** In 2025, nominal GDP (HK\$ 3,329,843 million) exceeds real GDP (HK\$ 3,301,474 million), reflecting cumulative positive price inflation relative to the benchmark year ($P_{2025} = 100.86$).

(2) Real GDP Growth Rates and Long-Term Average (5 points)

For each year $t \in [1962, 2025]$, the annual percentage growth rate of real GDP ($g_{Y,t}$) is given by:

$$g_{Y,t} = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \times 100\% \quad (1)$$

Computing the arithmetic mean across all 64 continuous annual growth rates (identical to Excel's AVERAGE function):

$$\bar{g}_Y = \frac{1}{64} \sum_{t=1962}^{2025} g_{Y,t} = \mathbf{5.45\%} \quad (5.4468\%) \quad (2)$$

Macroeconomic Highlights: Hong Kong exhibited rapid double-digit growth during early export industrialization, peaking at +16.16% in 1976. Major cyclical troughs reflect severe external shocks: −5.88% during the 1998 Asian Financial Crisis and an all-time low of −6.54% during the 2020 pandemic shock.

(3) GDP Deflator and Inflation Rates: Discrete vs. Logarithmic (5 points)

The implicit GDP deflator (P_t , base year 2024 = 100) and the discrete percentage inflation rate (π_t) are defined as:

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100, \quad \pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\% \quad (3)$$

Economists also frequently use the natural log-difference to approximate the instantaneous rate of inflation:

$$\pi_t^{\log} = \Delta \ln P_t \times 100\% = (\ln P_t - \ln P_{t-1}) \times 100\% \quad (4)$$

Both series are plotted concurrently in Figure 2.

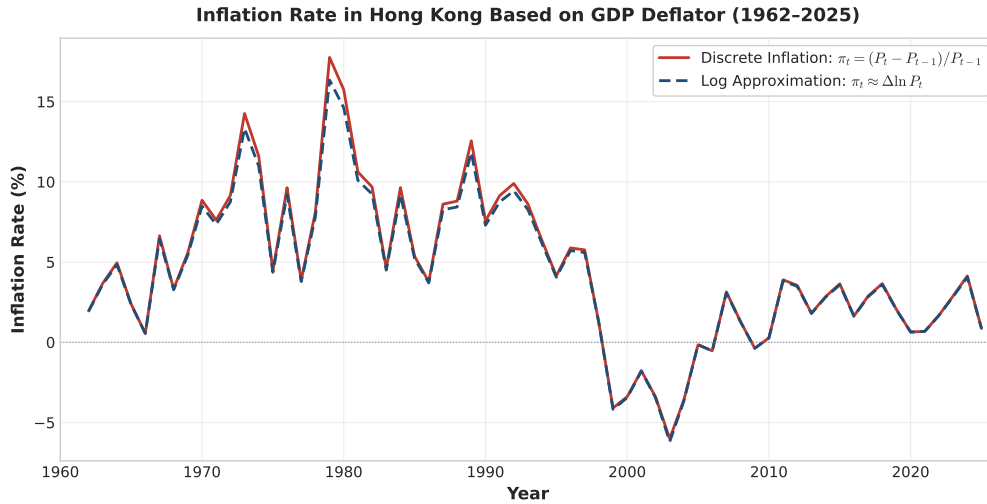


Figure 2: Comparison of Discrete and Logarithmic Inflation Rates (1962–2025)

Mathematical Foundation and Comparative Analysis:

- **High Empirical Co-movement:** The two series are almost indistinguishable over the 64-year sample, yielding a Pearson correlation coefficient $\rho = 0.9995$ and an average absolute discrepancy of merely 0.2035%.
- **Analytical Justification via Taylor Series:** Factoring the logarithmic difference:

$$\Delta \ln P_t = \ln \left(\frac{P_t}{P_{t-1}} \right) = \ln \left(1 + \frac{P_t - P_{t-1}}{P_{t-1}} \right) = \ln(1 + \pi_t) \quad (5)$$

Expanding $\ln(1 + x)$ via a first-order Maclaurin series around zero for small inflation rates ($|\pi_t| \ll 1$):

$$\ln(1 + \pi_t) = \pi_t - \frac{1}{2}\pi_t^2 + \frac{1}{3}\pi_t^3 - \dots = \pi_t - \frac{1}{2}\pi_t^2 + \mathcal{O}(\pi_t^3) \quad (6)$$

The approximation error is of order $\frac{1}{2}\pi_t^2$. When inflation is moderate ($\pi_t \approx 0$), higher-order terms vanish, proving $\Delta \ln P_t \approx \pi_t$.

- **Regime Divergence:** Observable separation occurs strictly in high-inflation regimes. In 1979, discrete inflation peaked at +17.88% while log-inflation yielded +16.45% (a 1.41% gap). Conversely, during low-inflation regimes (e.g., 2015–2025), both measures align. The series also captures Hong Kong’s protracted deflationary era from 1998 to 2004 ($\pi_t < 0$, troughing at −6.18% in 2003).

2 The Composition of HK GDP, 2023 (25 Points)

(4) Expenditure Components Accounting Table (10 points)

Using official data from Table 310-31002 (*Gross Domestic Product (GDP) by major expenditure component at current market prices*), the 2023 expenditure components are tabulated in Table 1. In modern national accounts, total gross investment (I) encompasses gross domestic fixed capital formation (I_{fixed}) and changes in inventories (ΔInv): $I = I_{\text{fixed}} + \Delta \text{Inv}$.

Table 1: Expenditure Components of Hong Kong GDP (2023, Current Market Prices)

Component	Millions of HKD, 2023	Percent of GDP (%)
Gross Domestic Product (Y)	2,981,210	100.00%
Consumption (C)	2,087,569	70.02%
Investment (I)	481,511	16.15%
<i>Gross domestic fixed capital formation</i>	<i>505,510</i>	<i>16.96%</i>
<i>Changes in inventories</i>	<i>-23,999</i>	<i>-0.81%</i>
Government spending (G)	396,813	13.31%
Net Exports (NX)	15,317	0.51%
Exports of goods and services (X)	5,272,395	176.85%
Imports of goods and services (IM)	5,257,078	176.34%
Inventory investment (ΔInv)	-23,999	-0.81%

National Accounting Identity Balance Check:

$$C + I + G + NX = 2,087,569 + 481,511 + 396,813 + 15,317 = 2,981,210 \equiv Y \quad (\text{Exact closure})$$

(5) Dominant Component of GDP (5 points)

Among the four primary components (C, I, G, NX), **Private Consumption Expenditure (C)** is the largest component, totaling **HK\$ 2,087,569 million** and accounting for **70.02%** of total GDP in 2023.

(6) Trade Openness Index and US Comparison (5 points)

From Table 310-31002, the trade and output statistics for Hong Kong in 2022 are:

$$\text{Exports of goods and services } (X_{2022}) = \text{HK\$ } 5,463,019 \text{ million} \quad (7)$$

$$\text{Imports of goods and services } (IM_{2022}) = \text{HK\$ } 5,348,126 \text{ million} \quad (8)$$

$$\text{Gross Domestic Product } (Y_{2022}) = \text{HK\$ } 2,808,922 \text{ million} \quad (9)$$

Evaluating the trade openness index:

$$\text{Openness}_{HK,2022} = \frac{X_{2022} + IM_{2022}}{Y_{2022}} = \frac{5,463,019 + 5,348,126}{2,808,922} = \frac{10,811,145}{2,808,922} \approx \mathbf{3.85} \quad (\mathbf{384.89\%}) \quad (10)$$

Economic Comparison with the US Economy (25% in 2023): Hong Kong's openness index of **384.89%** is more than **15 times larger** than that of the United States. This divergence reflects fundamental structural differences:

1. **Global Entrepôt and Re-export Intermediation:** Hong Kong operates as an open trading hub and logistics center where trade is measured in gross turnover (X, IM), while GDP measures domestic value-added (Y). Intermediate merchandise processed through Hong Kong easily exceeds domestic value-added multiple times over.
2. **Continental Economy vs. Small Open Economy:** The US is a self-contained, continental economy where internal interstate trade forms the primary economic base, meaning cross-border trade naturally constitutes a smaller fraction of aggregate GDP.

(7) Industry-Wise Decomposition (5 points)

From Table 310-34101 (*Gross Domestic Product (GDP) by major economic activity - percentage contribution to GDP*), the contributions of the five primary economic sectors in 2023 are:

1. Agriculture, fishing, mining and quarrying: 0.0%
2. Manufacturing: 1.0%
3. Electricity, gas and water supply, and waste management: 1.1%
4. Construction: 4.3%
5. **Services: 93.5%**

The most important industry is overwhelmingly **Services**, accounting for **93.5%** of Hong Kong's GDP in 2023. Key drivers include Financing and Insurance (24.9%) and Import/Export, Wholesale and Retail Trades (17.5%).

3 The Employment Rate (10 Points)

(8) Derivation of the Employment Rate (10 points)

Let the macroeconomic labor aggregates be defined as:

- WAP: Working-Age Population
- L : Labor Force, where $L = E + U$ (E : Employed, U : Unemployed)

Following standard OECD definitions:

$$\text{Participation Rate (PR)} \equiv \frac{L}{\text{WAP}} = 0.80 \quad (11)$$

$$\text{Unemployment Rate (u)} \equiv \frac{U}{L} = \frac{L - E}{L} = 1 - \frac{E}{L} = 0.10 \implies \frac{E}{L} = 1 - u = 0.90 \quad (12)$$

$$\text{Employment Rate (ER)} \equiv \frac{E}{\text{WAP}} \quad (13)$$

Factoring the employment rate definition:

$$ER = \frac{E}{\text{WAP}} = \left(\frac{E}{L}\right) \times \left(\frac{L}{\text{WAP}}\right) = (1 - u) \times PR \quad (14)$$

Substituting the parameter values:

$$ER = (1 - 0.10) \times 0.80 = 0.90 \times 0.80 = \mathbf{0.72} \quad (\mathbf{72.0\%}) \quad (15)$$

The employment rate in this economy is **72%**.

4 Goods Market Equilibrium and Multipliers (55 Points)

Consider the following goods market system for a closed economy (in billions of euros):

$$C = 480 + 0.5Y_D, \quad I = 110, \quad T = 70, \quad G = 250, \quad Y_D \equiv Y - T \quad (16)$$

(9) Goods Market Equilibrium Output (Y^*) (10 points)

In equilibrium, aggregate production equals aggregate demand ($Y = Z$):

$$Y = C + I + G = [480 + 0.5(Y - 70)] + 110 + 250 \quad (17)$$

$$Y = 480 + 0.5Y - 35 + 110 + 250 = 805 + 0.5Y \quad (18)$$

$$(1 - 0.5)Y = 805 \implies 0.5Y = 805 \implies Y^* = \frac{805}{0.5} = \mathbf{1,610} \text{ billion euros} \quad (19)$$

(10) Equilibrium Disposable Income (Y_D^*) (5 points)

$$Y_D^* = Y^* - T = 1,610 - 70 = \mathbf{1,540} \text{ billion euros} \quad (20)$$

(11) Equilibrium Consumption (C^*) (5 points)

$$C^* = 480 + 0.5(1,540) = 480 + 770 = \mathbf{1,250} \text{ billion euros} \quad (21)$$

Clearing Check: $Z^* = C^* + I + G = 1,250 + 110 + 250 = 1,610 \equiv Y^*$.

(12) Savings and Investment Accounting (5 points)

• **Private Savings (S_{private}):**

$$S_{\text{private}} = Y_D^* - C^* = 1,540 - 1,250 = \mathbf{290} \text{ billion euros} \quad (22)$$

• **Public Savings (S_{public}):**

$$S_{\text{public}} = T - G = 70 - 250 = \mathbf{-180} \text{ billion euros} \quad (\text{Budget Deficit of 180}) \quad (23)$$

• **Investment Spending (I):**

$$I = \mathbf{110} \text{ billion euros} \quad (24)$$

IS Identity Check: $S_{\text{total}} = S_{\text{private}} + S_{\text{public}} = 290 + (-180) = 110 \equiv I$.

(13) Government Spending Multiplier ($\Delta Y/\Delta G$) (5 points)

Holding taxes fixed ($\Delta T = 0$):

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - c_1} = \frac{1}{1 - 0.5} = \frac{1}{0.5} = \mathbf{2} \quad (25)$$

A one-unit expansion in G increases equilibrium output by **2 units**.

(14) Tax Multiplier ($\Delta Y/\Delta T$) (5 points)

Holding government spending fixed ($\Delta G = 0$):

$$\frac{\Delta Y}{\Delta T} = \frac{-c_1}{1 - c_1} = \frac{-0.5}{1 - 0.5} = \frac{-0.5}{0.5} = \mathbf{-1} \quad (26)$$

A one-unit increase in lump-sum taxes contracts equilibrium output by **1 unit**.

(15) Balanced Budget Multiplier ($\Delta Y/\Delta G|_{\Delta G=\Delta T}$) (5 points)

Under a balanced budget expansion ($\Delta G = \Delta T = 1$):

$$\left. \frac{\Delta Y}{\Delta G} \right|_{\Delta G=\Delta T} = \frac{\Delta Y}{\Delta G} + \frac{\Delta Y}{\Delta T} = \frac{1}{1 - c_1} + \frac{-c_1}{1 - c_1} = \frac{1 - c_1}{1 - c_1} = \mathbf{1} \quad (27)$$

Output expands by exactly the increase in government spending (Haavelmo theorem).

(16) Income-Dependent Taxes and Recalculated Multiplier (10 points)

Suppose taxes depend endogenously on aggregate income:

$$T = t_0 + t_1 Y = -252 + 0.2Y, \quad \text{where } t_0 = -252, t_1 = 0.2 \quad (28)$$

Substituting this tax rule into $Y = C + I + G$:

$$Y = c_0 + c_1[Y - (-252 + 0.2Y)] + I + G \quad (29)$$

$$Y = c_0 + c_1(1 - 0.2)Y - c_1(-252) + I + G \quad (30)$$

$$[1 - c_1(1 - t_1)]Y = [c_0 - c_1 t_0 + I + G] \implies Y = \frac{c_0 - c_1 t_0 + I + G}{1 - c_1(1 - t_1)} \quad (31)$$

Differentiating with respect to G :

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - c_1(1 - t_1)} = \frac{1}{1 - 0.5(1 - 0.2)} = \frac{1}{1 - 0.4} = \frac{1}{0.6} = \frac{5}{3} \approx 1.67 \quad (32)$$

The government spending multiplier falls from 2.00 to **1.67** (or $5/3$).

(17) Macroeconomic Rationale for Automatic Stabilizers (5 points)

An income-dependent tax system acts as an **automatic stabilizer** for two reasons:

1. **Built-in Cushioning of Disposable Income:** When an adverse demand shock contracts output ($\Delta Y < 0$), tax revenue automatically declines by $\Delta T = t_1 \Delta Y < 0$. This automatic tax relief cushions disposable income: $\Delta Y_D = (1 - t_1) \Delta Y$, meaning disposable income contracts less than output ($|\Delta Y_D| < |\Delta Y|$). Consequently, private consumption declines less sharply than under lump-sum taxes.
2. **Reduction of Multiplier and Volatility without Policy Lag:** The expenditure multiplier decreases from $\frac{1}{1 - c_1} = 2.0$ to $\frac{1}{1 - c_1(1 - t_1)} \approx 1.67$. Because this damping occurs instantaneously without legislative deliberation or implementation lag, it dampens cyclical volatility and stabilizes aggregate demand.